

Qualification Pack



Industrial wastewater Treatment Plant Operator

QP Code: PSC/Q0602

Version: 1.0

NSQF Level: 4

Water Management and Plumbing Skill Council || Unit No-606, DLF Prime Tower
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PSC/Q0602: Industrial wastewater Treatment Plant Operator

Brief Job Description

Industrial Wastewater Treatment Plant Operator responsible for operating and maintaining effluent treatment systems, monitoring water quality parameters, conducting routine tests, adjusting chemical dosages, troubleshooting equipment issues, and ensuring compliance with environmental regulations.

Personal Attributes

The individual must be able to work independently and be comfortable in performing laborious work. The person should be good at following instructions and a team player with a positive attitude.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSC/N0170: Operating Effluent Treatment Plant \(ETP\)](#)
2. [PSC/N0171: Monitoring and managing Effluent Treatment Plant](#)
3. [PSC/N0172: Analysis of Effluent](#)
4. [PSC/N0603: Maintain Records & Reporting at Site](#)
5. [PSC/N0602: Health and Safety Practices at the treatment Plant](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations, Plumbing Systems Installation and Maintenance
Country	India
NSQF Level	4
Credits	15

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/3132.0400
Minimum Educational Qualification & Experience	12th grade Pass OR Previous relevant Qualification of NSQF Level (NSQF Level 3.5) with 1.5 years of experience relevant experience in Water Supply/ Water Treatment/ Quality Control OR Previous relevant Qualification of NSQF Level (NSQF Level 3.0) with 3 Years of experience relevant experience in Water Supply/ Water Treatment/ Quality Control OR 11th Class with 1.5 years of experience relevant experience in Water Supply/ Water Treatment/ Quality Control
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	27/08/2027
NSQC Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QG-04-WS-02939-2024-V1-WMPS
NQR Version	1.0

Remarks:

NIL

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PSC/N0170: Operating Effluent Treatment Plant (ETP)

Description

This unit is about operating the Effluent treatment plant and understanding the various types of unit operations in effluent treatment plants. This unit is also about different terminologies that are used in ETP.

Scope

The scope covers the following :

- Prepare for effluent treatment
- Carry out operations in ETP

Elements and Performance Criteria

Prepare for Effluent treatment

To be competent, the user/individual on the job must be able to:

- PC1.** Start the plant by switching on the pump, mixer, controllers, and other equipment.
- PC2.** Perform the manual work in the effluent treatment plant.
- PC3.** Check, open, or close the valves.
- PC4.** Perform the operations at outflow & inflow of the plant by controlling the discharge.
- PC5.** Collect the samples and facilitate the calibrations of samples.

Carry out operations in ETP

To be competent, the user/individual on the job must be able to:

- PC6.** Removal of large solids such as rags, sticks, grit, and grease that may damage equipment.
- PC7.** Removal of floating and settled materials such as suspended solids and organic matter.
- PC8.** Removal of biodegradable organic matter and suspended solids.
- PC9.** Perform the Aerobic biological treatment technologies.
- PC10.** Perform biotreatment process by Conventional Activated Sludge Process (CASP).
- PC11.** Can carry out another biotreatment process by Cyclic Activated Sludge System (CASS).
- PC12.** Familiar with how to carry out Integrated Fixed Film Activated Sludge (IFAS) system.
- PC13.** Knowledge of the process of Membrane Bioreactor (MBR) for biological degradation.
- PC14.** Removal of residual suspended solids/dissolved solids.
- PC15.** Complete the documentation work as required.

Operational control Strategies

To be competent, the user/individual on the job must be able to:

- PC16.** Removal of wastewater by mixed liquor suspended solids process.
- PC17.** Calculation of Sludge volume index and sludge density index.
- PC18.** Sludge age: Mean Cell Residence Time (MCRT).
- PC19.** Able to calculate the Food/Mass Ratio.
- PC20.** Able to calculate Constant MLSS.

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PC21. Perform Return Activated Sludge Control (RAS).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance and need of treating the water.
- KU2.** Basic terminologies like discharge
- KU3.** Physical characteristics of water like total solids, suspended solids, settled solids, etc.
- KU4.** Chemical characteristics of water like ph chloride content, nitrogen content, fats, oil & grease, sulfates, hydrogen sulphide, dissolved oxygen, etc.
- KU5.** Biological characteristics of water like Bod ,Cod, MLSS,pH etc.
- KU6.** Decomposition by aerobic, anaerobic & facultative bacteria, neutralization, absorption, adsorption, oxidation/reduction, ion exchange process.
- KU7.** Disposal of Effluents.
- KU8.** Biological operations like sludge digestion
- KU9.** Process of removal of floating, settleable organic matter and suspended solids.
- KU10.** Method to clean different units present in the effluent treatment plant.
- KU11.** Handling of chemicals.
- KU12.** Process of effluent treatment performed during treatment of chemical waste.
- KU13.** Different methods of handling the effluent.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, report and fill out forms in local language or Hindi or English.
- GS2.** Understand the various color codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** ability to read from different sources- books screens in machines and signage.
- GS4.** Perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** Read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS6.** State information, doubts and concerns about work related matters in local language or Hindi/English.
- GS7.** Plan ones dailly tasks to achieve maximum efficiency.
- GS8.** Be punctual and work as per agreed priorities
- GS9.** Manage distractions and maintain workplace discipline
- GS10.** Listen to the higher authorities carefully.
- GS11.** be courteous
- GS12.** Identify ways to increase efficiency and reduce errors.
- GS13.** Read and understand manuals, health and safety instructions, memos, other company documents.



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GS14. Communicate with supervisor.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for Effluent treatment</i>	7	14	-	2
PC1. Start the plant by switching on the pump, mixer, controllers, and other equipment.	-	-	-	-
PC2. Perform the manual work in the effluent treatment plant.	-	-	-	-
PC3. Check, open, or close the valves.	-	-	-	-
PC4. Perform the operations at outflow & inflow of the plant by controlling the discharge.	-	-	-	-
PC5. Collect the samples and facilitate the calibrations of samples.	-	-	-	-
<i>Carry out operations in ETP</i>	15	30	-	5
PC6. Removal of large solids such as rags, sticks, grit, and grease that may damage equipment.	-	-	-	-
PC7. Removal of floating and settled materials such as suspended solids and organic matter.	-	-	-	-
PC8. Removal of biodegradable organic matter and suspended solids.	-	-	-	-
PC9. Perform the Aerobic biological treatment technologies.	-	-	-	-
PC10. Perform biotreatment process by Conventional Activated Sludge Process (CASP).	-	-	-	-
PC11. Can carry out another biotreatment process by Cyclic Activated Sludge System (CASS).	-	-	-	-
PC12. Familiar with how to carry out Integrated Fixed Film Activated Sludge (IFAS) system.	-	-	-	-
PC13. Knowledge of the process of Membrane Bioreactor (MBR) for biological degradation.	-	-	-	-
PC14. Removal of residual suspended solids/dissolved solids.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Complete the documentation work as required.	-	-	-	-
<i>Operational control Strategies</i>	8	16	-	3
PC16. Removal of wastewater by mixed liquor suspended solids process.	-	-	-	-
PC17. Calculation of Sludge volume index and sludge density index.	-	-	-	-
PC18. Sludge age: Mean Cell Residence Time (MCRT).	-	-	-	-
PC19. Able to calculate the Food/Mass Ratio.	-	-	-	-
PC20. Able to calculate Constant MLSS.	-	-	-	-
PC21. Perform Return Activated Sludge Control (RAS).	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0170
NOS Name	Operating Effluent Treatment Plant (ETP)
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0171: Monitoring and managing Effluent Treatment Plant

Description

This unit is based on monitoring and managing the effluent treatment plant and to ensure proper working of effluent treatment plant and how the problems should be troubleshoot in case of any discrepancy.

Scope

The scope covers the following :

- Monitoring the ETP
- Managing the ETP
- Troubleshooting the problems in ETP

Elements and Performance Criteria

Monitoring the ETP

To be competent, the user/individual on the job must be able to:

- PC1.** Monitor the working of mechanical equipment of ETP
- PC2.** proper check of mechanical instruments such as blowers, pumps, centrifugal separators, valves, etc
- PC3.** Monitor the physical quality of water at the intermediate of treatment and at outflow of the plant.
- PC4.** Monitor weather the secondary treatment is carried out as per the standards.
- PC5.** Proper check the meter and gauge readings in the plant.
- PC6.** Proper check the working of all the treatment units that carries out treatment.
- PC7.** Evaluate the performance of biological treatment processes, such as activated sludge, biofilters, or lagoons
- PC8.** Monitor the biomass concentration, sludge age, and microbial activity to ensure proper treatment. Adjust aeration and nutrient dosing if required.

Managing the ETP

To be competent, the user/individual on the job must be able to:

- PC9.** Measure and monitor the flow rates of wastewater and treatment chemicals throughout the system
- PC10.** Regularly test the influent and effluent water to assess its quality and identify any deviations from the required standards
- PC11.** Clean or repair these components to maintain proper sedimentation and clarification processes.
- PC12.** Check the condition of filters, such as sand filters, activated carbon filters, or membrane filters
- PC13.** Monitor sludge quality, volume, and disposal practices according to regulatory requirements.

Troubleshooting the problems in ETP

To be competent, the user/individual on the job must be able to:

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- PC14.** Check all physical components, such as pumps, valves, pipes, and filters, for any signs of damage, leaks, or blockages.
- PC15.** Repair or replace faulty components as necessary.
- PC16.** Examine settling tanks and clarifiers for sludge accumulation, improper sludge blanket levels, or mechanical issues
- PC17.** Clean or repair these components to maintain proper sedimentation and clarification processes.
- PC18.** Verify that all operating parameters, such as temperature, pressure, pH, and retention times, are within the recommended ranges and if not then take standard measures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different operations performed in treatment of Effluent like Screening, Grit Chamber, Flow Equalization Chamber, Detritus Tank, Aeration Tank, Skimming Tank, Primary Sedimentation Tank, Coagulation aided Sedimentation, Sludge digestion, Trickling Filters, Rotatory Biological Contractors, Membrane Bio Reactors, Moving Bed Bio Reactors, Septic Tank, Activated Sludge Process, Secondary Sedimentation Tank, Sludge Drying Beds, 3rd degree treatment.
- KU2.** Process of purifying gas in Scrubber Unit like Purifying gas evolved during sludge digestion process.
- KU3.** Knowledge of all mechanical equipment in ETP
- KU4.** Method to clean different units present in Effluent Treatment Plant.
- KU5.** Handling of chemicals.
- KU6.** Process flow of Effluent treatment in the plant.
- KU7.** Different methods of handling the sludge.
- KU8.** Types of filters in ETP
- KU9.** Able to read meter and gauge reading

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** Ability to read from different sources- books screens in machines and signage.
- GS4.** Perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** Read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS6.** State information, doubts and concerns about work related matters in local language or Hindi/English
- GS7.** Plan one's daily tasks to achieve maximum efficiency
- GS8.** Be punctual and work as per agreed priorities



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- GS9.** Manage distractions and maintain workplace discipline
- GS10.** Listen to the higher authorities carefully
- GS11.** Be courteous
- GS12.** Identify ways to increase efficiency and reduce errors
- GS13.** Read and understand manuals, health and safety instructions, memos, other company documents.
- GS14.** Communicate with supervisor.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Monitoring the ETP</i>	10	20	-	3
PC1. Monitor the working of mechanical equipment of ETP	-	-	-	-
PC2. proper check of mechanical instruments such as blowers, pumps, centrifugal separators, valves, etc	-	-	-	-
PC3. Monitor the physical quality of water at the intermediate of treatment and at outflow of the plant.	-	-	-	-
PC4. Monitor whether the secondary treatment is carried out as per the standards.	-	-	-	-
PC5. Proper check the meter and gauge readings in the plant.	-	-	-	-
PC6. Proper check the working of all the treatment units that carries out treatment.	-	-	-	-
PC7. Evaluate the performance of biological treatment processes, such as activated sludge, biofilters, or lagoons	-	-	-	-
PC8. Monitor the biomass concentration, sludge age, and microbial activity to ensure proper treatment. Adjust aeration and nutrient dosing if required.	-	-	-	-
<i>Managing the ETP</i>	10	20	-	4
PC9. Measure and monitor the flow rates of wastewater and treatment chemicals throughout the system	-	-	-	-
PC10. Regularly test the influent and effluent water to assess its quality and identify any deviations from the required standards	-	-	-	-
PC11. Clean or repair these components to maintain proper sedimentation and clarification processes.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Check the condition of filters, such as sand filters, activated carbon filters, or membrane filters	-	-	-	-
PC13. Monitor sludge quality, volume, and disposal practices according to regulatory requirements.	-	-	-	-
<i>Troubleshooting the problems in ETP</i>	10	20	-	3
PC14. Check all physical components, such as pumps, valves, pipes, and filters, for any signs of damage, leaks, or blockages.	-	-	-	-
PC15. Repair or replace faulty components as necessary.	-	-	-	-
PC16. Examine settling tanks and clarifiers for sludge accumulation, improper sludge blanket levels, or mechanical issues	-	-	-	-
PC17. Clean or repair these components to maintain proper sedimentation and clarification processes.	-	-	-	-
PC18. Verify that all operating parameters, such as temperature, pressure, pH, and retention times, are within the recommended ranges and if not take standard measures.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0171
NOS Name	Monitoring and managing Effluent Treatment Plant
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0172: Analysis of Effluent

Description

This unit is based on the analysis of physical, chemical, biological quality of effluent.

Scope

The scope covers the following :

- identify the commonly used chemicals.
- Perform chemical tests according to procedures.
- Precautions during testing

Elements and Performance Criteria

identify the commonly used chemicals

To be competent, the user/individual on the job must be able to:

- PC1.** Familiar with the chemicals used in Coagulation and flocculation like acrylamide copolymers, aluminium chloride, aluminium sulphate, etc...
- PC2.** An individual should be aware of chemicals used in pH adjustment like calcium carbonate, calcium hydroxide, calcium oxide, carbon dioxide, etc...
- PC3.** Chemicals used in disinfection and oxidation products such as anhydrous ammonia, ammonium hydroxide, calcium hypochlorite, etc...
- PC4.** Different chemicals used in softening, Algicide, Taste and odour control, Dechlorinate and antioxidant, Bleaches, etc...
- PC5.** And individual should know the use and applications with some other major chemicals like Ferric alum or Ammonium iron (III) sulphate, Lime (calcium oxide), Polyelectrolyte (flocculant) and De-chlorinator

Perform chemical tests according to procedures

To be competent, the user/individual on the job must be able to:

- PC6.** Able to perform Chemical Oxygen Demand (COD)
- PC7.** Analyse Microbial test
- PC8.** An individual must be able to carry out Dissolved Oxygen Test (Winkler Method)
- PC9.** An individual should be able to carry out Biochemical Oxygen Demand (BOD)
- PC10.** Total Suspended Solids (TSS)
- PC11.** Total Dissolved Solids (TDS)
- PC12.** Oil & Grease Content
- PC13.** Jar Testing
- PC14.** Other relevant tests

Precautions during testing

To be competent, the user/individual on the job must be able to:

- PC15.** Make sure that all chemicals in the laboratory have proper labels.
- PC16.** Perform the use of chemicals, gases, or reagents with labels.

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- PC17.** Perform the mixing of chemicals or reagents with the information at hand about their reactivity.
- PC18.** Individual should be familiar with proper Personal Protective Equipment (PPE) for handling chemicals in the laboratory.
- PC19.** Ensure that the individual should hold only updated MSDS.
- PC20.** Make sure chemical emergency equipment such as an eye washer and a safety shower are available
- PC21.** Perform the proper disposal of used chemicals or reagents in the different container as per the guidelines.
- PC22.** An individual should avoid storing chemicals over his/her requirement in the laboratory as per SOP.
- PC23.** An individual should avoid chemicals from heat source or direct sunshine.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Reagents used in different tests
- KU2.** Standard operating Procedure for carrying out test
- KU3.** General principles of the tests
- KU4.** Step by step process of carrying out experiment
- KU5.** Use of PPE kits in different conditions
- KU6.** Storage of chemicals in different conditions following standard operating procedure.
- KU7.** Reactivity of chemicals
- KU8.** Suitable temperature for different types of chemicals at which they must be stored.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** Ability to read from different sources- books screens in machines and signage.
- GS4.** Perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** Read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS6.** State information, doubts and concerns about work related matters in local language or Hindi/English
- GS7.** Plan one's daily tasks to achieve maximum efficiency
- GS8.** Be punctual and work as per agreed priorities
- GS9.** Manage distractions and maintain workplace discipline
- GS10.** Listen to the higher authorities carefully



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- GS11.** Be courteous
- GS12.** Identify ways to increase efficiency and reduce errors
- GS13.** Read and understand manuals, health and safety instructions, memos, other company documents.
- GS14.** Communicate with supervisor.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify the commonly used chemicals</i>	8	16	-	2
PC1. Familiar with the chemicals used in Coagulation and flocculation like acrylamide copolymers, aluminium chloride, aluminium sulphate, etc...	-	-	-	-
PC2. An individual should be aware of chemicals used in pH adjustment like calcium carbonate, calcium hydroxide, calcium oxide, carbon dioxide, etc...	-	-	-	-
PC3. Chemicals used in disinfection and oxidation products such as anhydrous ammonia, ammonium hydroxide, calcium hypochlorite, etc...	-	-	-	-
PC4. Different chemicals used in softening, Algicide, Taste and odour control, Dechlorinate and antioxidant, Bleaches, etc...	-	-	-	-
PC5. An individual should know the use and applications with some other major chemicals like Ferric alum or Ammonium iron (III) sulphate, Lime (calcium oxide), Polyelectrolyte (flocculant) and De-chlorinator	-	-	-	-
<i>Perform chemical tests according to procedures</i>	15	30	-	4
PC6. Able to perform Chemical Oxygen Demand (COD)	-	-	-	-
PC7. Analyse Microbial test	-	-	-	-
PC8. An individual must be able to carry out Dissolved Oxygen Test (Winkler Method)	-	-	-	-
PC9. An individual should be able to carry out Biochemical Oxygen Demand (BOD)	-	-	-	-
PC10. Total Suspended Solids (TSS)	-	-	-	-
PC11. Total Dissolved Solids (TDS)	-	-	-	-
PC12. Oil & Grease Content	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Jar Testing	-	-	-	-
PC14. Other relevant tests	-	-	-	-
<i>Precautions during testing</i>	7	14	-	4
PC15. Make sure that all chemicals in the laboratory have proper labels.	-	-	-	-
PC16. Perform the use of chemicals, gases, or reagents with labels.	-	-	-	-
PC17. Perform the mixing of chemicals or reagents with the information at hand about their reactivity.	-	-	-	-
PC18. Individual should be familiar with proper Personal Protective Equipment (PPE) for handling chemicals in the laboratory.	-	-	-	-
PC19. Ensure that the individual should hold only updated MSDS.	-	-	-	-
PC20. Make sure chemical emergency equipment such as an eye washer and a safety shower are available	-	-	-	-
PC21. Perform the proper disposal of used chemicals or reagents in the different container as per the guidelines.	-	-	-	-
PC22. An individual should avoid storing chemicals over his/her requirement in the laboratory as per SOP.	-	-	-	-
PC23. An individual should avoid chemicals from heat source or direct sunshine.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0172
NOS Name	Analysis of Effluent
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	3.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0603: Maintain Records & Reporting at Site

Description

This unit focuses on the importance of maintaining accurate records and effectively reporting activities at a sewage treatment plant. It emphasizes the significance of documentation in ensuring compliance with regulations, monitoring plant performance, and facilitating continuous improvement. Trainees will learn how to record data, prepare reports, and communicate relevant information to stakeholders.

Scope

The scope covers the following :

- Understand how to maintain records and data
- Reporting to the higher officials
- Respond to higher officials

Elements and Performance Criteria

Understand how to maintain records and data

To be competent, the user/individual on the job must be able to:

- PC1.** able to read plan & specifications for different types of modifications in sewage treatment plant.
- PC2.** able to document data on pathogen reduction.
- PC3.** able to record the data on sludge accumulated and sludge treated.
- PC4.** able to maintain daily operational log.
- PC5.** perform routine operational duties, flow meter readings, sample information with corresponding analytical data, etc.
- PC6.** able to identify unusual conditions at the site
- PC7.** identify and report accident equipment hazard.
- PC8.** perform preventive maintenance

Reporting to the higher officials

To be competent, the user/individual on the job must be able to:

- PC9.** reporting to the supervisor if any kind of problem occurs
- PC10.** identify spare parts inventory when in need
- PC11.** identify specifications on equipment from supplier
- PC12.** equipment inventory

Respond to higher officials

To be competent, the user/individual on the job must be able to:

- PC13.** able to responding promptly to higher officials' inquiries or requests demonstrates a sense of urgency and professionalism.
- PC14.** able to provide clear and accurate information in responses to higher officials ensures effective communication and minimizes misunderstandings.

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- PC15.** able to ensure that responses align with relevant policies, procedures, and regulations demonstrates a commitment to adhering to organizational guidelines.
- PC16.** able to maintain respectful and courteous tone in responses conveys professionalism and enhances the organization's reputation.
- PC17.** identify potential solutions or recommendations in responses demonstrates proactive thinking and a commitment to finding effective resolutions.
- PC18.** able to follow up on requests or inquiries to higher officials and providing updates or additional information as needed helps to maintain effective communication and ensure that all concerns are addressed satisfactorily.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand local, state, and national regulations related to sewage treatment, including record-keeping and reporting requirements mandated by environmental agencies
- KU2.** Familiarity with different types of records, such as daily log sheets, maintenance logs, chemical usage reports, effluent quality records, energy consumption, and equipment performance logs.
- KU3.** Knowledge of how to sample influent and effluent water, record results from tests (e.g., pH, BOD, COD, TSS), and ensure these meet environment
- KU4.** Knowledge of keeping detailed logs of routine and emergency maintenance activities on equipment (e.g., pumps, aerators, clarifiers) to track equipment lifespan, repairs, and prevent failures.
- KU5.** Understand protocols for reporting spills, overflows, equipment malfunctions, or non-compliance events to regulatory authorities as required.
- KU6.** Knowledge of SOPs for different tasks within the sewage treatment plant, and the importance of maintaining updated records that align with these procedures.
- KU7.** Documenting the scheduled preventive maintenance tasks and keeping track of their completion to minimize downtime and optimize plant performance.
- KU8.** Knowledge to record data related to odor, noise, and emissions (if applicable) to ensure the plants environmental impact is minimized and within regulatory limits.
- KU9.** Understand the use of digital systems for data logging, report generation, and data retrieval. These systems may include SCADA (Supervisory Control and Data Acquisition) or other monitoring tools.
- KU10.** Generate performance reports for internal review, management oversight, and submission to regulatory bodies.
- KU11.** Knowledge of proper archiving of records, both in digital and hard copy formats, in accordance with legal retention policies, and ensuring easy retrieval during audits.
- KU12.** Knowledge of documentation procedures during emergencies (e.g., system failures, flooding) and having templates for swift reporting in urgent situations.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** ability to read from different sources- books screens in machines and signage.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or hindi
- GS5.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS6.** listen to the higher authorities carefully
- GS7.** communicate with supervisor.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand how to maintain records and data</i>	10	20	-	3
PC1. able to read plan & specifications for different types of modifications in sewage treatment plant.	-	-	-	-
PC2. able to document data on pathogen reduction.	-	-	-	-
PC3. able to record the data on sludge accumulated and sludge treated.	-	-	-	-
PC4. able to maintain daily operational log.	-	-	-	-
PC5. perform routine operational duties, flow meter readings, sample information with corresponding analytical data, etc.	-	-	-	-
PC6. able to identify unusual conditions at the site	-	-	-	-
PC7. identify and report accident equipment hazard.	-	-	-	-
PC8. perform preventive maintenance	-	-	-	-
<i>Reporting to the higher officials</i>	10	20	-	4
PC9. reporting to the supervisor if any kind of problem occurs	-	-	-	-
PC10. identify spare parts inventory when in need	-	-	-	-
PC11. identify specifications on equipment from supplier	-	-	-	-
PC12. equipment inventory	-	-	-	-
<i>Respond to higher officials</i>	10	20	-	3
PC13. able to responding promptly to higher officials' inquiries or requests demonstrates a sense of urgency and professionalism.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. able to provide clear and accurate information in responses to higher officials ensures effective communication and minimizes misunderstandings.	-	-	-	-
PC15. able to ensure that responses align with relevant policies, procedures, and regulations demonstrates a commitment to adhering to organizational guidelines.	-	-	-	-
PC16. able to maintain respectful and courteous tone in responses conveys professionalism and enhances the organization's reputation.	-	-	-	-
PC17. identify potential solutions or recommendations in responses demonstrates proactive thinking and a commitment to finding effective resolutions.	-	-	-	-
PC18. able to follow up on requests or inquiries to higher officials and providing updates or additional information as needed helps to maintain effective communication and ensure that all concerns are addressed satisfactorily.	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0603
NOS Name	Maintain Records & Reporting at Site
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0602: Health and Safety Practices at the treatment Plant

Description

this unit is about the health-related issues at stp and safety practices which workers need to be taken care off. health and safety practices plays a crucial role to ensure the well-being and protection of employees, visitors, and the environment. these practices aim to minimize the risks associated with handling and treating wastewater, hazardous chemicals, and potential biological contaminants.

Scope

The scope covers the following :

- Maintaining health at treatment plant
- safe work practises at the Treatment facilities

Elements and Performance Criteria

maintaining health at treatment plant

To be competent, the user/individual on the job must be able to:

- PC1.** identify the hazards associated with different tasks in a wastewater treatment plant.
- PC2.** able to understand and acquire know-how of safety measures to be taken when in project site.
- PC3.** identify general risks, including working at height and hazards associated with different tasks in a sewage treatment plant.
- PC4.** understand and acquire know-how of safety measures when handling mechanical parts in statutory or moving position.
- PC5.** perform the changing of chlorine cylinders safely.
- PC6.** perform the automatic and manual functions at scrubber unit safely as per instructions.

safe work practices at treatment facilities

To be competent, the user/individual on the job must be able to:

- PC7.** inform the higher authorities of the accident or hazard that has occurred.
- PC8.** select the right personal protective equipment for different tasks in a waste treatment plant.
- PC9.** ensure safe storage of corrosive substances.
- PC10.** follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with
- PC11.** able to clean and disinfect work area, materials/supplies, equipment etc.. before and after use.
- PC12.** able to identify hygiene and sanitation issues.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** possible causes of risk, hazard or accident in the workplace

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- KU2.** location of all the general health and safety equipment in the workplace
- KU3.** work practices and precautions to control and prevent risks, hazards and accidents
- KU4.** meaning of hazards and risks
- KU5.** do and donts in operation of waste treatment plant
- KU6.** practise of safe storage of corrosive substances.
- KU7.** able to detect chlorine leakage and control, scrubber unit gas leakage and control

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS5.** plan one's daily tasks to achieve maximum efficiency
- GS6.** be punctual and work as per agreed priorities
- GS7.** manage distractions and maintain workplace discipline
- GS8.** listen to the higher authorities carefully
- GS9.** be courteous
- GS10.** read and understand manuals, health and safety instructions, memos, other company documents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>maintaining health at treatment plant</i>	15	30	-	5
PC1. identify the hazards associated with different tasks in a wastewater treatment plant.	-	-	-	-
PC2. able to understand and acquire know-how of safety measures to be taken when in project site.	-	-	-	-
PC3. identify general risks, including working at height and hazards associated with different tasks in a sewage treatment plant.	-	-	-	-
PC4. understand and acquire know-how of safety measures when handling mechanical parts in statutory or moving position.	-	-	-	-
PC5. perform the changing of chlorine cylinders safely.	-	-	-	-
PC6. perform the automatic and manual functions at scrubber unit safely as per instructions.	-	-	-	-
<i>safe work practices at treatment facilities</i>	15	30	-	5
PC7. inform the higher authorities of the accident or hazard that has occurred.	-	-	-	-
PC8. select the right personal protective equipment for different tasks in a waste treatment plant.	-	-	-	-
PC9. ensure safe storage of corrosive substances.	-	-	-	-
PC10. follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with	-	-	-	-
PC11. able to clean and disinfect work area, materials/supplies, equipment etc.. before and after use.	-	-	-	-
PC12. able to identify hygiene and sanitation issues.	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0602
NOS Name	Health and Safety Practices at the treatment Plant
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0170.Operating Effluent Treatment Plant (ETP)	30	60	0	10	100	25
PSC/N0171.Monitoring and managing Effluent Treatment Plant	30	60	0	10	100	20
PSC/N0172.Analysis of Effluent	30	60	0	10	100	25
PSC/N0603.Maintain Records & Reporting at Site	30	60	0	10	100	15
PSC/N0602.Health and Safety Practices at the treatment Plant	30	60	0	10	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	170	330	-	50	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
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Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements that together specify the technical, generic, professional and organisational specific knowledge that an individual need in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of those OS, these include communication related skills that are applicable to most job roles.
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Options	Options are NOS or set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.